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1. Execute the Coursera Lab Jupyter notebook for this lesson, which implements the Monte-Carlo method to approximate an integral using both the standard method and the importance-density method. This is the same example presented in the lesson.

1 / 1 point

What is the root-mean-squared error between the true integral value and the estimate produced by the (Monte-Carlo with) importance-density method over the final 2000 iterations? Enter your answer with four digits to the right of the decimal point.

Note: Make sure that the code is configured with the default values of `N=10000` and `sx = 0.75` if you have changed them to experiment. Also make sure that the code executes `randn("state", 0)` before generating Gaussian random samples for the Monte-Carlo method and that it executes `rand("state", 1)` before generating uniform random samples for the (Monte-Carlo with) importance-density method; otherwise the result of running the code will not agree with the expectations of the quiz auto-grader.

0.0023

✔ Correct

Yes. That is correct. Interestingly, the RMSE is better than for the standard Monte-Carlo method in this case.

2. For the same Coursera Lab Jupyter notebook, change the value of `sx` from the default of `0.75` to the new value of `1.0`. Do not modify anything else from the default conditions.

1 / 1 point

What is the root-mean-squared error between the true integral value and the estimate produced by the (Monte-Carlo with) importance-density method over the final 2000 iterations? Enter your answer with four digits to the right of the decimal point.

0.0027

✔ Correct

Yes, that is correct.

3. For the same Coursera Lab Jupyter notebook, change the value of `N` from the default of `10000` to the new value of `25000`. Do not modify anything else from the default conditions. (Make sure that you have changed `sx` back to the default value of `0.75`, for example.)

1 / 1 point

What is the root-mean-squared error between the true integral value and the estimate produced by the (Monte-Carlo with) importance-density method over the final 2000 iterations? Enter your answer with four digits to the right of the decimal point.

0.0074

✔ Correct

Yes, that is correct. Interestingly, using more samples gives a worse result in this case.